UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE

SEMESTER II - 2023/2024

MAT 1023 - REAL ANALYSIS I

MID SEMESTER EXAMINATION

Time allowed: ONE hour **Answer ALL questions**

Instructions : Provide a clear, logically structured justification to earn the allocated marks.

1. **(20 points)** Show that $(p \implies r) \ \wedge \ (q \implies r)$ and $(p \vee q) \implies r$ are logically equivalent. (You may use a truth table or standard equivalence laws.)

[Turn over...

2. **(24 points)** Suppose $x, y \in \mathbb{R}$. For each statement, prove it or give a counterexample, with justification. (You may use the fact that the set of rational numbers is closed under addition and multiplication.)

   (a) If x is irrational and y is irrational, then $x + y$ is irrational.

   (b) If $x + y$ is irrational, then x is irrational or y is irrational.

   (c) If x is irrational and y is irrational, then xy is irrational.

   (d) If xy is irrational, then x is irrational or y is irrational.

3. **(16 points)** Let $a, b \in \mathbb{R}$. By clearly stating the relevant **Field Axioms**, prove that

$$(-a) \cdot (-b) = a \cdot b$$

4. **(16 points)** Let $x, y \in \mathbb{R}$ such that $0 < x < y$. By clearly stating the relevant **Order Axioms**, prove that

$$0 < \frac{1}{y} < \frac{1}{x}$$

[Turn over...

5. **(24 points)** For each set S, determine $\inf S,\ \sup S,\ \min S,\ \max S$ (if they exist), with justification.

(a) $S_1 = \{1, \pi/3, \pi^2, 10, e\}$

(b) $S_2 = \left\{ \sin\left(\frac{n\pi}{3}\right) : n \in \mathbb{Z}^+ \cup \{0\} \right\}$

(c) $S_3 = (-2024, 2025)$

(d) $S_4 = \{m/n \in \mathbb{Q} \ : \ m^2 < 5n^2 \text{ and } m, n > 0\}$